

Reviewer Pack — Minimal Geometric Langlands Dimension Correlation with Resol...

Entry df51ab9d90c5 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Geometric Langlands Dimension Correlation with Resolution Proof Width

Entry ID: df51ab9d90c5 **Verdict:** INCONCLUSIVE **Recorded:** 2026-06-01 14:52:19 UTC

1. Conjecture

Field A (mathematical branch): Geometric Langlands Theory **Field B** (complexity object): Resolution Proof Complexity

Statement:

For every CNF φ with n variables, the minimal geometric Langlands dimension ($\text{gld}(\varphi)$) of its associated vector space is linearly correlated with its resolution proof width $w(\varphi)$, such that $\text{gld}(\varphi) = \Theta(w(\varphi))$.

Rationale (proposer's reasoning):

The Geometric Langlands program provides a bridge between algebraic geometry and number theory, which may reveal new insights into the structure of computational complexity. A correlation between geometric Langlands dimensions and proof complexity could indicate a deep connection between these fields.

Taxonomy category: geometric_langlands (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 0d42e4e9c42b9b31

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if the Pearson correlation coefficient between $\text{gld}(\varphi)$ and $w(\varphi)$ exceeds 0.7 for all CNFs φ with $n \leq 40$ across at least 30 random seeds, AND no seed has a correlation coefficient below 0.5.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: ADJACENT_OK against 1 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - "Geometric Langlands Theory" AND "resolution proof complexity" AND "minimal geometric langlands dimension" - " $\text{gld}(\phi)$ " AND "resolution proof width $w(\phi)$ " - " $\theta(w(\phi))$ correlation Geometric Langlands Theory resolution proof complexity"

Top relevant hits considered: - [s2:ca44ef9be97a1c211295785ba4b0290db90a6bf2] international workshop on functions in algebra and geometry 2nd-6th 2022

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math
from fractions import Fraction

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_cnf(n):
        cnf = []
        for _ in range(n):
            clause = [random.randint(1, n), -random.randint(1, n)]
            cnf.append(clause)
        return cnf

    def resolution_width(cnf):
        clauses = set(tuple(sorted(c)) for c in cnf)
        learned_clauses = []

        while True:
            new_clause = None
            for i in range(len(clauses)):
                for j in range(i + 1, len(clauses)):
                    a, b = clauses[i], clauses[j]
                    if -a[0] in b and -a[1] in b:
                        new_clause = tuple(sorted([-b[0], -b[1]]))
                        break
                if new_clause:
                    break
            if not new_clause:
                return len(learned_clauses)
            learned_clauses.append(new_clause)
```

```

        clauses.add(new_clause)

def geometric_langlands_dimension(cnf):
    # Placeholder for actual computation; returns a random value for testing
    return random.uniform(1, 10)

n_values = [5, 10, 15, 20, 30, 40]
gld_values = []
w_values = []

for n in n_values:
    cnf = generate_cnf(n)
    gld = geometric_langlands_dimension(cnf)
    w = resolution_width(cnf)

    if not (gld is not None and w is not None):
        return {
            "metric_name": "correlation_coefficient",
            "metric_value": 0.0,
            "instances_tested": len(n_values),
            "n_max": n,
            "conjecture_holds": False,
            "counterexample": "mapping_undefined"
        }

    gld_values.append(gld)
    w_values.append(w)

mean_gld = sum(gld_values) / len(gld_values)
mean_w = sum(w_values) / len(w_values)

correlation_coefficient = sum((gld - mean_gld) * (w - mean_w) for gld, w in zip(gld_values, w_values))

return {
    "metric_name": "correlation_coefficient",
    "metric_value": correlation_coefficient,
    "instances_tested": len(n_values),
    "n_max": max(n_values),
    "conjecture_holds": correlation_coefficient >= 0.7 and all(cc >= 0.5 for cc in [correlation_coefficient]),
    "counterexample": ""
}

if __name__ == "__main__":
    import sys
    seeds = list(map(int, sys.argv[1:])) or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29] * 3

    results = []

    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_metric_value = sum(r["metric_value"] for r in results) / len(results)
    std_metric_value = math.sqrt(sum((r["metric_value"] - mean_metric_value) ** 2 for r in results) / len(results))
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

```

```

if all(r["conjecture_holds"] for r in results):
    print(f"RESULT: SUPPORTED mean={mean_metric_value} std={std_metric_value} support_fraction={support_fraction}")
elif any(not r["conjecture_holds"] and r["metric_value"] < 0.5 for r in results):
    first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"] and r["metric_value"] < 0.5)
    print(f"RESULT: FALSIFIED counterexample=\"low_correlation\" first_failing_seed={first_failing_seed}")
else:
    print(f"RESULT: INCONCLUSIVE support_fraction={support_fraction} n_tested={len(results)}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7d9c9605.py", line 94, in <module>
    result = run_trial(seed)
              ~~~~~^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7d9c9605.py", line 58, in run_trial
    w = resolution_width(cnf)
        ~~~~~^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7d9c9605.py", line 36, in resolution_width
    a, b = clauses[i], clauses[j]
            ~~~~~^
TypeError: 'set' object is not subscriptable

```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test code crashed before producing data, which prevents us from evaluating the conjecture's support conditions. | next: Investigate and fix the error in the test code to ensure it can complete without crashing.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	15886
2	preregistration	ollama_remote	glm4:latest	0	0	10321
3	novelty	ollama_remote	glm4:latest	0	0	8642
4	novelty	ollama_remote	glm4:latest	0	0	9167
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10676

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	18683
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	16387
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	16588
9	judge	ollama_remote	glm4:latest	0	0	9132

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 115482 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in research/audit/df51ab9d90c5.jsonl)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in research/audit/df51ab9d90c5.jsonl for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: research/pvsnp_sandbox/test_*.py (per-cycle)

Replay tarball: research/replay/df51ab9d90c5.tar.gz (if generated)